



Report

DES

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Step 1

Convert the Text and Key to hexadecimal then to binary

Text: Mohmed77

Key: 43170285

After Convert to hexadecimal

Text: 0x4D6F686D65643737

Key: 0x3433313730323835

After Convert hexadecimal to binary

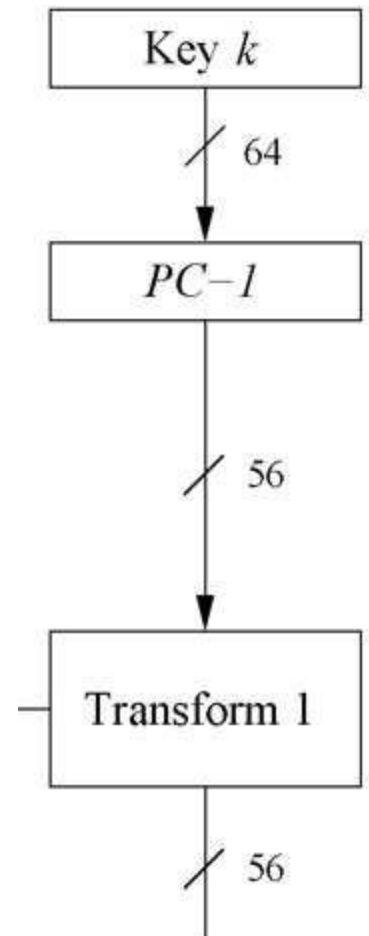
Text:01001101011011110110100001101101011001010110010000110111
00110111

Key:001101000011001100110001001101110011000000110010001110000
0110101

Step 2

Permute place the binary key using the PC-1 schedule.

$PC - 1$							
57	49	41	33	25	17	9	1
58	50	42	34	26	18	10	2
59	51	43	35	27	19	11	3
60	52	44	36	63	55	47	39
31	23	15	7	62	54	46	38
30	22	14	6	61	53	45	37
29	21	13	5	28	20	12	4



Key after permuting 56 bit:

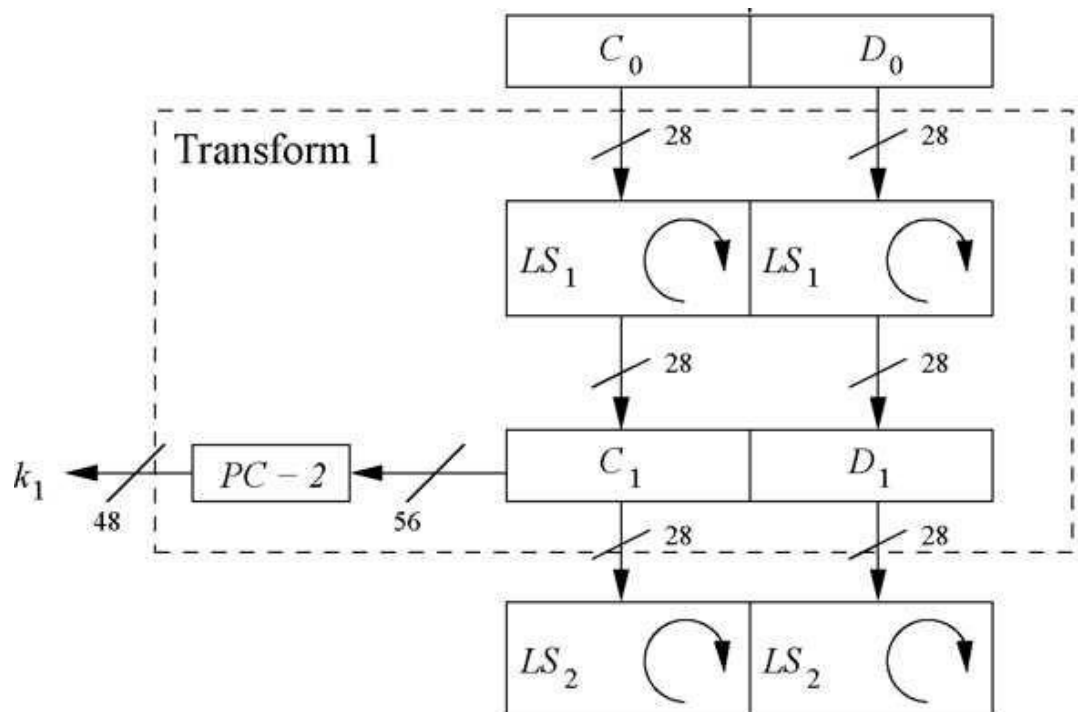
00000000000000001111111111101010000000100110000000
01111

Step 3

Now we divide the key after permuting into two parts.

$C_0 = 0000000000000000111111111111$

$D_0 = 0101000000010011100000001111$



Then we will do Left rotate to C and D.

- In Rounds 1,2,9, 16 Rotate 1 bit.

- in other Rounds rotate 2 bit. ---> 3,4,5,6,7,8,10,11,12,13,14,15

Let's start to rotate :)

$C_1 = 000000000000000011111111111110$

$D_1 = 1010000000100110000000011110$

$C_2 = 000000000000000011111111111100$

$D_2 = 0100000001001100000000111101$

$C_3 = 0000000000001111111111110000$

$D_3 = 0000000100110000000011110101$

$C_4 = 0000000000111111111111000000$

$D_4 = 0000010011000000001111010100$

$C_5 = 0000000011111111111100000000$

$D_5 = 0001001100000000111101010000$

$C_6 = 0000001111111111110000000000$

$D_6 = 0100110000000011110101000000$

$C_7 = 0000111111111111000000000000$

$D_7 = 0011000000001111010100000001$

$C_8 = 0011111111111100000000000000$

$D_8 = 1100000000111101010000000100$

$C_9 = 0111111111111100000000000000$

$D_9 = 1000000001111010100000001001$

$C_{10} = 11111111111100000000000000001$

$D_{10} = 0000000111101010000000100110$

$C_{11} = 11111111100000000000000000111$

$D_{11} = 0000011110101000000010011000$

$C_{12} = 11111110000000000000000011111$

$D_{12} = 0001111010100000001001100000$

$C_{13} = 11111000000000000000001111111$

$D_{13} = 0111101010000000100110000000$

$C_{14} = 1110000000000000000111111111$

$D_{14} = 1110101000000010011000000001$

$C_{15} = 1000000000000000001111111111$

$D_{15} = 1010100000001001100000000111$

$C_{16} = 0000000000000000011111111111$

$D_{16} = 0101000000010011000000001111$

Step 4

Now put the C and D together, then Permuting place every binary key using the PC-2 schedule.

CD₁: 000000000000000111111111110101000000010011000
0000011110

CD₂: 00000000000000011111111111000100000001001100000
000111101

CD₃: 0000000000000001111111111100000000000100110000000
011110101

CD₄: 00000000000111111111110000000000010011000000001
111010100

CD₅: 00000000111111111111000000000001001100000000111
101010000

CD₆: 00000011111111111100000000000100110000000011110
101000000

CD₇: 00001111111111110000000000000011000000001111010
100000001

CD₈: 0011111111111100000000000000110000000011110101
0000000100

CD₉: 011111111111110000000000000001000000001111010100
000001001

CD₁₀: 11111111111100000000000000001000000011110101000
0000100110

CD₁₁: 11111111100000000000000011100000111010100000
0010011

CD₁₂: 1111111000000000000000001111100011101010000000
1001100000

CD₁₃: 1111100000000000000000111111011110101000000010
0110000000

CD₁₄: 111000000000000000001111111111010100000001001
1000000001

CD₁₅: 100000000000000000111111111101010000000100110
0000000111

CD₁₆: 0000000000000000001111111111010100000001001100
000000111000

Now we Permuting place every binary key using the PC-2 schedule, the key will be 48 bit.

<i>PC – 2</i>							
14	17	11	24	1	5	3	28
15	6	21	10	23	19	12	4
26	8	16	7	27	20	13	2
41	52	31	37	47	55	30	40
51	45	33	48	44	49	39	56
34	53	46	42	50	36	29	32

Key 1:

010100000010110010101100011001000000001001010010

Key 2:

010100001010110010100100110000101000000101010000

Key 3:

110100001010110000100110010000011000011100001100

Key 4:

111000001010011000100110010110000001010010001000

Key 5:

111000001001011000100110010010000101000000101101

Key 6:

111000001001001001110010000000100111100010101000

Key 7:

101001001101001001110010101000000001100100110001

Key 8:

101001100101001101010010100000110000101000110010

Key 9:

001001100101001101010011100000010100001101000010

Key 10:

001011110101000101010001100101001000001000000100

Key 11:

000011110100000111011001110100000000011011000100

Key 12:

000111110100000110011001000110001010001010001001

Key 13:

000111110000100110001001001100100111010000000001

Key 14:

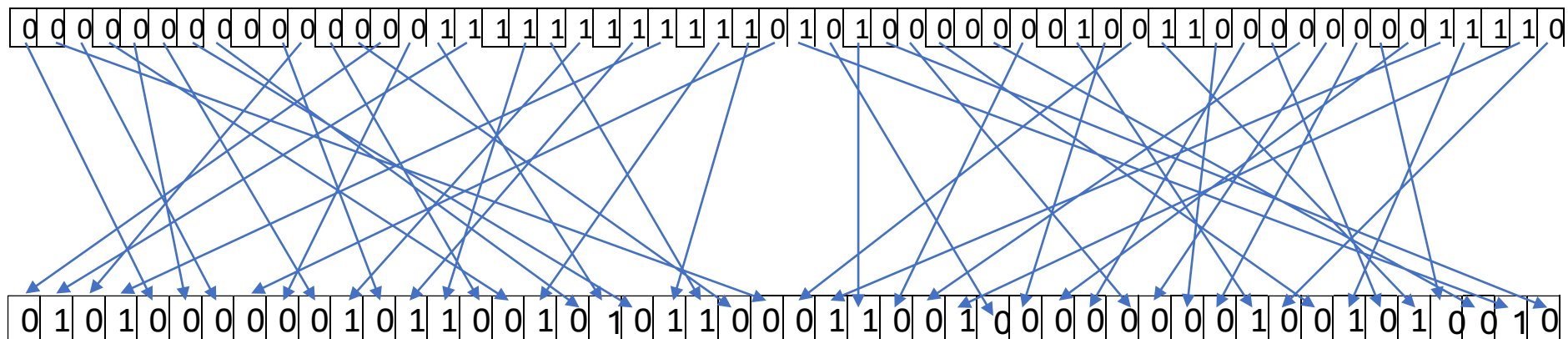
000110110010100010001101001010100010000100100010

Key 15:

000110010010110010001100101001000110100100000010

Key 16:

010100010010110010001100000001110000100101000001



This Example how to get key 1 from CD₁, using PC-2 permuting Schedule.

Step 5

Now we use the initial schedule to Permuting the binary text.

Initial Permutation

<i>IP</i>							
58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

Original Text Binary:

010010000110000101100010011011100100011001100001011
0100101110011

After Permuting:

111111111000000000011000111000100000000011101110010
0100110011100

Step 6

Here we will divide the binary text into two parts, L (Left) and R (Right), Left part will do a XOR with f(function), Right part will take two copy first copy go to the Left part in the second round, second copy go to the f(function) with the key.

The Expression: $L_i = R_{i-1}$

$$R_i = L_{i-1} \oplus f(R_{i-1}, k_i)$$

Text After Permuting by Initial Schedule:

111111111000000000011000111000100000000011101110010
0100110011100

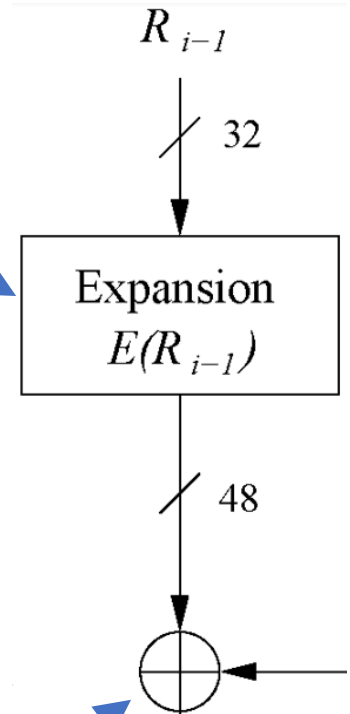
L_0 : 11111111100000000001100011100010

R_0 : 00000000111011100100100110011100

Step 7

now we are in the function, The right part of binary text is 32 bit and the key is 48 bits, so we must let the right part of binary text 48 bits, so we will use the Expansion $E(R_{i-1})$.

E					
32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

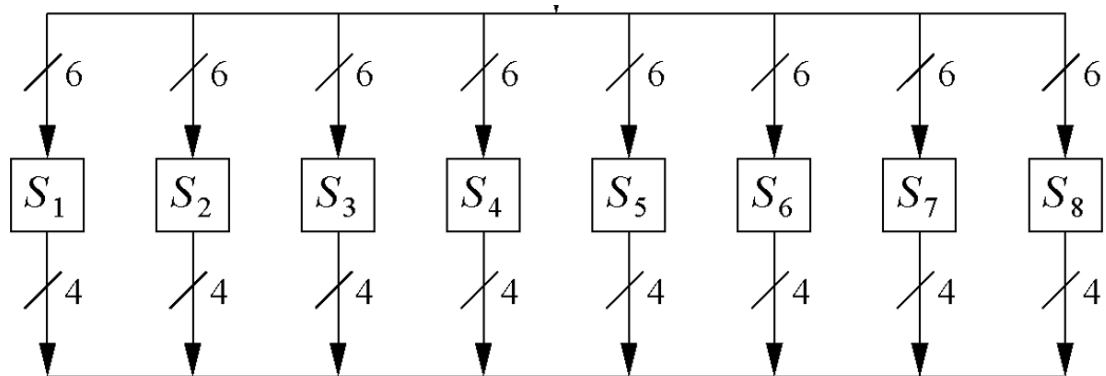


Step 8

After Expansion $E(R_{i-1})$, now do XOR between the output of Expansion and the key that we got it.

Step 9

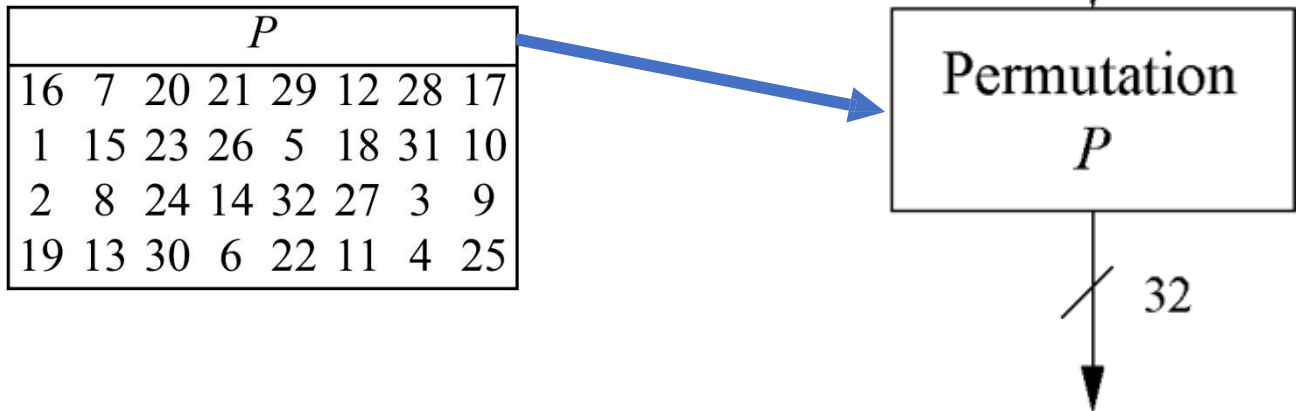
After XOR now go to the last step in function it's s-box, after XOR you have 48 bit, so it will divide to 8 boxes, 6 bits for each box and the output will be 4 bits for each box.



	S_1															
0	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
1	0	15	7	3	14	2	13	1	10	6	12	11	9	5	3	8
2	4	1	14	8	13	6	2	11	15	12	9	7	13	10	5	0
3	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13
	S_2															
0	15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
1	3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
2	0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
3	13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9
	S_3															
0	10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
1	13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
2	13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
3	1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12
	S_4															
0	7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
1	13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
2	10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
3	3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14
	S_5															
0	2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
1	14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
2	4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
3	11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3
	S_6															
0	12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
1	10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
2	9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
3	4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13
	S_7															
0	4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
1	13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
2	1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
3	6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12
	S_8															
0	13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
1	1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
2	7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
3	2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

Step 10

Now after s-box do the permutation using the schedule.



Step 11

After get the output from s-box now do XOR between L_0 and $f(\text{function})$, then output will be R in second round.

Note: I'll do the steps together step by step.

Round 1

L₀: 1111111110000000001100011100010

R₀: 00000000111011100100100110011100

R₀ Before Expansion:

00000000111011100100100110011100

R₀ After Expansion:

00000000000101110101110000100101001111001111

1000

R₀ $\oplus$ Key₁: 010100000011101111110000010000010011111010101010

S₁: 010100 --> Row: 00 = 0 , Column: 1010 = 10 | = 6 ---> 0110

S₂: 000011 --> Row: 01 = 1 , Column: 0001 = 1 | = 13 ---> 1101

S₃: 101111 --> Row: 11 = 3 , Column: 0111 = 7 | = 7 ---> 0111

S₄: 110000 --> Row: 10 = 2 , Column: 1000 = 8 | = 15 ---> 1111

S₅: 010000 --> Row: 00 = 0 , Column: 1000 = 8 | = 8 ---> 1000

S₆: 010011 --> Row: 01 = 1 , Column: 1001 = 9 | = 1 ---> 0001

S₇: 111010 --> Row: 10 = 2 , Column: 1101 = 13 | = 5 ---> 0101

S₈: 101010 --> Row: 10 = 2 , Column: 0101 = 5 | = 12 ---> 1100

Output from s-box: 01101101011111111000000101011100

After permutation: 10001111010110011111001001110100

f(function)= 10001111010110011111001001110100

L₀ $\oplus$ f(function) = 01110000110110011110101010010110

L₁ = R₀----- > **Next round**

R₁ = L₀ $\oplus$ f(function) ----- > **Next round**

	S_1																
0	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7	
1	0	15	7	3	14	2	13	1	10	6	12	11	9	5	3	8	
2	4	1	14	8	13	6	2	11	15	12	9	7	13	10	5	0	
3	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13	
	S_2																
0	15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10	
1	3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5	
2	0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15	
3	13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9	
	S_3																
0	10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8	
1	13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1	
2	13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7	
3	1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12	
	S_4																
0	7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15	
1	13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9	
2	10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4	
3	3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14	
	S_5																
0	2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9	
1	14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6	
2	4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14	
3	11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3	
	S_6																
0	12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11	
1	10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8	
2	9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6	
3	4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13	
	S_7																
0	4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1	
1	13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6	
2	1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2	
3	6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12	
	S_8																
0	13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7	
1	1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2	
2	7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8	
3	2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11	

Round 2

L1: 00000000111011100100100110011100

R1: 01110000110110011110101010010110

R1 Before Expansion:

01110000110110011110101010010110

R1 After Expansion:

00111010000101101111001111101010101010010101100

R1 $\oplus$ Key2: 0110101010111010010101110011011111010101111110

S1: 011010 --> Row: 00 = 0 , Column: 1101 = 13 | = 9 ---> 1001

S2: 101011 --> Row: 11 = 3 , Column: 0101 = 5 | = 15 ---> 1111

S3: 101001 --> Row: 11 = 3 , Column: 0100 = 4 | = 6 ---> 0110

S4: 010111 --> Row: 01 = 1 , Column: 1011 = 11 | = 12 ---> 1100

S5: 001101 --> Row: 01 = 1 , Column: 0110 = 6 | = 13 ---> 1101

S6: 111101 --> Row: 11 = 3 , Column: 1110 = 14 | = 8 ---> 1000

S7: 010111 --> Row: 01 = 1 , Column: 1011 = 11 | = 12 ---> 1100

S8: 111100 --> Row: 10 = 2 , Column: 1110 = 14 | = 5 ---> 0101

Output from s-box: 10011111011011001101100011000101

After permutation: 01110001100111010101100001110111

f(function)= 01110001100111010101100001110111

L1 $\oplus$ f(function) = 01110001011100110001000111101011

L2 = R1 ----- > **Next round**

R2 = L1 $\oplus$ f(function) ----- > **Next round**

	S_1															
0	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
1	0	15	7	3	14	2	13	1	10	6	12	11	9	5	3	8
2	4	1	14	8	13	6	2	11	15	12	9	7	13	10	5	0
3	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13
	S_2															
0	15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
1	3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
2	0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
3	13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9
	S_3															
0	10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
1	13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
2	13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
3	1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12
	S_4															
0	7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
1	13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
2	10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
3	3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14
	S_5															
0	2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
1	14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
2	4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
3	11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3
	S_6															
0	12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
1	10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
2	9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
3	4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13
	S_7															
0	4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
1	13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
2	1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
3	6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12
	S_8															
0	13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
1	1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
2	7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
3	2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

Step 13

After repeating the round 1-16, Now we have L_{16} & R_{16} , after that we put R_{16} & L_{16} together.

Step 14

Here we do permutation to RL_{16} using Final Permutation Schedule.

IP^{-1}							
40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

